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MILITARY SPECIFICATION
TEST STATION, RADIO FREQUENCY
OJ-213( )/AWM-23(V)

This specification has been approved by the Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers the performance, design, and test requirements for the Radio Frequency Test Station (RFTS) OJ-213( )/AWM-23(V) (see 6.1).

1.2 Classification. The RFTS consists of a 4-bay station and accessories as follows" (see 6.4):

a. Bay No. 1: Console, Test, Radio Frequency (RF), Transmitter OJ-247( )/AWM-23(V)

	Nomenclature	Reference Paragraph
(1)	Panel , Test, Noise Measurement ID-1860( )/AWM-23(V)	3.6.1.1
(2)	RF Power Meter	3.6.1.2
(3)	RF voltmeter	3.6.1.3
(4)	Panel, Coolant Disconnect MX-9305( )/AWM-23(V)	3.6.1.4
(5)	Mounting MT-4498( )/AWM-23(V) (see 6.9)	3.6.1.5
(6)	Drawer, Coolant Drain (see 6.9)	3.6,1.6
(7)	Panel , Slave Access	3.6.1 7
(8)	Power Supply PP-6770( )/AWG-9	3.6.1.7.1
(9)	Power Supply PP-6771( )/AWG-9	3.6.1.7.2
(10)	Power Supply PP-6772( )/AWG-9	3.6.1.7.3

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to:

Engineering Specifications and Standards Department (Code 93) Naval Air Engineering Center, Lakehurst, NJ 08733, by using self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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b. Bay No. 2: Console, Test, Radio Frequency, Receiver OJ-248( )/AWM-23(V)

(1) Microwave Swept Oscillator	3.6.2.1
(2) Noise Figure Meter	3.6.2.2
(3) Spectrum Analyzer	3.6.2.3
(4) Switching Unit, Fault Isolation SA-1893( )/AWM-23(V)	3.6.2.4
(5) Test Set, Radio Frequency Power TS-3276( )/AWM-23(V)	3.6.2.5
(6) Oscillator, Radio Frequency 0-1633( )/AWG-9	3 6.2.5.1

c. Bay No. 3. Console, Test, Radio Frequency , Receiver-Transmitter OJ-249( )/AWM-23(V)

(1} Oscilloscope	3.6.3.1
(2) Panel, Test, Electrical C-9177( )/AWM-23(V)	3.6.3.2
(3) Panel, Access	3.6 3.3
(4) Comparator, Signal CM-434( )/AWM-23(V)	3.6.3.4
(5) Dummy Load, Electrical DA-630( )/AWM-23(V)	3-6.3.5
(6) Transducer TR-283( )/AWM-23(V)	3.6.3.6

d. Bay No. 4: Console, Test, Radio Frequency OJ-250( )/AWM-23(V)

(1) Panel , Power Distribution SB-3694( )/AWM-23(V)	3.6.4.1
(2) Signal Generator, RF	3.6.4.2
(3) Counter/Timer	3.6.4.3
(4) Frequency Counter	3.6.4.4
(s) Digital Multimeter	3.6.4.5
(6) Frequency Response Analyzer	3.6.4-6
(7) Range Simulator	3.6.4.7
(8) Generator, Reference Signal SG-1014( )/AWM-23(V)	3.6.4.8
(9) Power Supply PP-6848( )/AWM-23(V)	3.6.4.9
(10) Power Supply, DC	3.6.4.10

e. Accessories - RFTS 3.6.5

(1) Plug-in Modules	3.6.2.3
(2) Adapters, Mounting	3.6.5.1
(3) Mounting Adapter-Modulator	3.6.5.1.1
(4) Mounting Adapter-RMO and Receiver	3.6.5 .1.2
(5) Mounting Adapter-Power Supply	3.6.5.1.3
(6) Waveguides, Interface	3.6.5.2
(7) Extender Kit-RFTS	3.6.5.3
(8) Commercial Items, Auxiliary	3.6.5.4
(9) Fixture, Pressure Leakage-Test	--

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1.3 Associated Equipment. The RFTS shall operate with the following associated equipment (see 6.4).

a. Kit, Tool, Special TK-222( )/AWM-23(V)
b. Cooler, Liquid, Electronic Equipment HD-957( )/AWM-23(V)
c. Stand, Fill-Drain, Coolant System MT-4730( )/AWM-23(V)

2. APPLICABLE DOCUMENTS

2.1 General. The following documents, of the issue specified in AS-4053, Applicable Document List for AWCS Test Set AN/AWM-23(V), form a part of this specification to the extent specified herein. MIL-T-81734(AS) shall be the applicable general specification for the RFTS described herein (see 6.6).

SPECIFICATIONS

Military

MIL-M-7793	Meter, Time Totalizing
MIL-T-81734(AS)	Test Set, Airborne Weapon Control System, AN/AWM-23(V)

STANDARDS

Military

MIL-STD-461	Electromagnetic Interference Characteristics, Requirements for Equipment

PUBLICATIONS

Hughes Aircraft Company

3282204	Acceptance Test Procedure, RF Test Station

2.2 Availability of documents. Applicable (see 6.6).

2.3 Precedence of documents. When the requirements of the contract, this specification, or applicable subsidiary documents are in conflict, the following precedence shall apply:

a. Contract - The contract shall have precedence over any other document.
b. This specification - This specification shall have precedence over all applicable subsidiary documents (see 6.6).
c. Referenced documents - Any document referenced in this specification shall have precedence over all applicable documents referenced therein.

3. REQUIREMENTS

3.1 Materials, parts, and processes. Applicable (see 6.6).

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3.2 Design and construction. Appli cable except as follows.

3.2.1 Design features. Applicable.

3.2.1.1 Method of operation. Applicable.

3.2.1.2 Commercial test equipment. Applicable.

3.2.1.2.1 Other commercial components. Applicable.

5.2.1.3 Circuit grounds. Applicable.

3.2.1.4 Interchangeability. Applicable.

3.2.1.5 Identification of product. Applicable.

3.2.1.6 Human factors. Applicable.

3.2.1.7 Test set generated acoustic noise. Applicable.

3.2.1.8 Forced-liquid cooling. Applicable. The liquid coolant flow rate for the RFTS shall be 6.8 gallons per minute, maximum. The pressure shall be 175 pounds per square inch, maximum.

3.2.1.9 Space and weight limitations. The RFTS shall occupy less than 42 square feet of deck area. The weight shall be less than 3400 pounds. Maxi - mum dimensions shall be 97 inches by 54 inches by 75 inches with all drawers in the retracted position, and the mounting surface attached.

3.2.1.9.1 Storage space. All items which are not permanently attached to the RFTS will require separate storage. A minimum of 20 cubic feet of storage space will be required to store Weapon Replaceable Assembly (WRA) mounting adapters, test adapters, waveguides, cables, and manuals.

3.2.1.10 Connectors. Applicable.

3.2.1.11 Secondary electrical power. The total secondary power shall be limited to 1600 watts maximum with any combination of power form loading within the load capacity limits for each power form listed in Table I.

3.2.1.11.1 Secondary power, semi-regulated. Secondary electrical power forms shall be provided by power supplies which are part of the RFTS. Power forms from 10 percent to full load and for normal values of line voltage variations shall be as listed in Table I.

3.2.1.11.2 Secondary power, regulated. Secondary regulated power forms shall be provided by power supplies which are part of the RFTS. These voltages are used by the RFTS internally only. Regulation from no load to full load, and for normal values of line voltage shall not exceed 2 percent. Ripple voltages for dc power forms shall not exceed 150 millivolts (mV) peak-to-peak (p-p). The power forms listed in Table II shall be provided.

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TABLE I

SEMI-REGULATED SECONDARY POWER FORMS

Item	Power Form (Vdc)	Load Capacity (Amperes)
1	$+57^{+5.7}_{-4.5}$	6.0
2	$+31^{+5.1}_{-3.4}$	23.0
3	$+17^{+2.8}_{-1.8}$	33.0
4	$+9.7^{+1.6}_{-1.8}$	45.0
5	$-17^{-2.8}_{+1.8}$	18.0
6	$-31^{-5.1}_{+3.4}$	16.0
7	$-57^{-5.7}_{+4.5}$	4.0
8	$-162^{-16.0}_{+19.5}$	0.16

TABLE II

REGULATED SECONDARY POWER FORMS

Item	Power Form (Vdc)	Load Capacity (Amperes)*
1	+5	20.0
2	+12	1.6
3	+25	3.0
4	-12	0.6
5	-25	0.5

*The regulated secondary power load currents are part of the load capacity of 3.2.1.11.1

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3.2.1.12 Interface. The RFTS shall provide the following interfaces.

3.2.1.12.1 Liquid cooler interface. The RFTS shall interface with the liquid cooler through coolant lines which enter the top of bay No. 1.

3.2.1.12.2 Clean air interface. The RFTS shall interface with the clean air through ducts which enter the top rear of each console of the RFTS.

3.2-3.12.3 Electrical power interface. The RFTS shall interface with electrical power for convenience outlets at one end of the convenience outlet strip mounted at the top of the front of the RFTS. The RFTS shall interface with the 400 Hz power form through power lines which enter the top of bay no. 1. The RFTS shall interface with all other power forms through power lines which enter the top of bay no. 4.

3.2.2 Utilities. Applicable.

3.2.2.1 Electrical power. Applicable. Electrical power required for operation of the RFTS shall be as listed in Table III. The 115/200 Volt (V), 400 Hertz (Hz), 3-phase leads entering the RFTS will be a separate bus not shared with any other service from the facility source regulator.

TABLE III

RFTS ELECTRICAL POWER

Item	Input Power Form	Maximum Load
1	+29 ± 4 Vdc	0.77 kilowatts (kW)
2	115/200 Vac, 400 Hz, 3-phase, 4-wire, Type III	26 kilovolt amperes (kVA)
3	115 Vac, 60 Hz, 1-phase, Type I	1.75 kVA
4	115 Vac, 60 Hz, 3-phase 3-wire, ungrounded, Type I	4.2 kVA

3.2.2.2 Clean air. Applicable. The RFTS requirements for clean air shall not exceed 77 pounds per minute.

3.2.2.3 Potable water. Not applicable.

3.2.2.4 Liquid coolant. Applicable,

3.2.2.5 Helium. Applicable.

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3.2.2.6 Nitrogen. Applicable.

3.2.2.7 Hydraulic fluid. Not applicable.

3.2.3 Service conditions.

3.2.3.1 Electrical. Applicable.

2.3.2 Environmental. Applicable.

3.2.3.2.1 Organizational level. Not applicable.

3.2.3.2.2 Intermediate level. Applicable.

3.2.4 Maintainability. Applicable.

3.2.4.1 Mean time to repair. Applicable.

3.2.4.2 Test points. Applicable. Test points shall be provided to facilitate fault isolation to the shop replaceable assembly (SRA) level, fault isolation within an SRA, alignments or adjustments, calibration, and signal monitoring.

3.2.4.3 Shop replaceable assemblies- Applicable.

3.2.4.4 Self test capability. In the event of self test failure, the RFTS shall provide capability to establish its functional readiness. Time for self test shall not exceed 30 minutes. The RFTS shall provide capability to isolate faults to:

a. Commercial test equipment (CTE) item.
b. SRA level for panel and drawer assemblies.

3.2.5 Reliability. The mean time between failures shall be 60 hours minimum.

3.2.5.1 Operating life. Applicable.

3.2.5.2 Storage life. Applicable.

3.2.5.3 Operational stability. Applicable.

3.2.6 Electromagnetic interference (EMI). Applicable with the following exceptions:

a. MIL-STD-461, test CE03 limits for the 115/200 Vat, 400 Hz leads shall be 40 dB above the limits specified in Figures 14 and 15 of MIL-STD-461.
b. MIL-STD-461, test RS02 shall not apply to the 115/200 Vat, 400 Hz leads.
c. MIL-STD-461, tests CE03 and RS02-1 and the limits therein shall apply unless otherwise specified above.

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d. MIL-STD-461, test CE04 shall apply only when power remote control leads external to the RFTS are used.
e. MIL-STD-461, tests CE01, CE02, RS01, RS02-2 and RS04 shall not apply.

3.2.7 Utilities protection. Applicable.

3.2.7.1 Liquid cooling protection. Applicable, except that protection shall be provided only for an out of tolerance flow rate.

3.2.7.2 Clean air protection. Applicable.

3.2.7.3 Electrical power protection. Applicable.

3.2.7.3.1 Secondary voltage protection. Occurrence of an out-of-tolerance secondary voltage condition that could damage either an item under test or the test station shall result in automatic shutdown of the secondary power forms. The out-of-tolerance condition leading to shutdown shall be identified by an appropriate indicator. Operator action shall be required to reinstate test station operation after the occurrence of automatic shutdown.

3.2.7.3.2 Emergency shutdown switch. An emergency shutdown switch shall be provided for use by the operator. When an emergency shutdown switch is activated, the load side of all primary circuit breakers shall be disconnected from their respective loads. The switch utilized for this application shall be of the pushbutton-activated, momentary type, with the pushbutton finished in red. The front panel area around the pushbutton shall be marked with alternate orange-yellow and black stripes. The switch shall be-located on the RFTS, in an area which is easily accessible to the operator during normal operation.

3.2.8 Calibration cycle. Applicable.

3.2.9 Power switching and coolant control. The RFTS shall provide the necessary switching to control utility electrical power forms and shall provide valves and baffling for control of coolants.

3.2.10 Displays. Displays shall provide an indication of the RFTS test status.

3.3 Performance.

3.3.1 General. The RFTS shall provide the capability to test and maintain the equipment listed in Table IV.

3.3.2 Organizational level functional requirements. Not applicable.

3.3.3 Intermediate level functional requirements. Applicable.

3.3.4 Warmup time. Applicable.

3.4 Detailed functional requirements.

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3.4.1 Test capabilities. The RFTS shall provide and control the stimuli, power forms, and the loads required to perform functional and diagnostic testing of the WRAs listed in Table IV. Test capabilities shall be as specified in 3.4.1.1 through 3.4.1.9.

TABLE IV

WRAs MAINTAINED AT THE RF TEST STATION

Item	WRA Nomenclature (see 6.4)
1	Oscillator, Radio Frequency O-1633( )/AWG-9
2	Synchronizer-Transmitter SN-487( )/AWG-9
3	Transmitter, Radar T-1224( )/AWG-9
4	Power Supply PP-6770( )/AWG-9
5	Power Supply PP-6771( )/AWG-9
6	Power Supply PP-6772( )/AWG-9
7	Receiver, Radar R-1785( )/AWG-9
8	Converter, Signal Data CV-2950( )/AWG-9 and CV-3225( )/AWG-9
9	Converter, Signal Data CV-2946( )/AWG-9, CV-3224( )/AWG-9 and CV-3366( )/AWG-9

3.4.1.1 Oscillator, Radio Frequency 0-1633( )/AWG-9. The RFTS shall provide the capability for performing the following tests on the radio frequency oscillator.

a. Local oscillator output

(1) Phase lock. Measure the transmitter oscillator/local oscillator (TO/LO) offset frequency to within ± 10 Hz of the nominal intermediate frequency (IF).
(2) Power. Measure LO power output to an accuracy of ± 1.0 dB in the region of +6 dB referred to one milliwatt (dBm).
(3) Frequency. Measure LO frequency to an accuracy of ± 1 kHz at X-band.
(4) Noise. Measure the amplitude modulation (AM) noise to an accuracy of ± 3 dB in the range 100 to 140 dB below the carrier.

b. Transmitter oscillator (TO) output

(1) Frequency. Measure the TO frequency to an accuracy of ± 1 kHz at X-band.
(2) Power. Measure the TO power output to an accuracy of ± 0.6 dB in the region of +17 dBm.

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(3) Frequency modulated ranging (FMR) linearity. Measure frequency variation from the ideal FMR slope characteristics, in the frequency region about the ideal frequency, to an accuracy of ± 50 Hz.
(4) AIM-54 missile rear link command. - Measure frequency to an accuracy of ± 1 kHz in the X-band region.
(5) AIM-7F pulse doppler modulation. Measure:

(a) Modulation frequency to an accuracy of ± 0.5 Hz.
(b) Modulation index to an accuracy of ± 5 percent for indices between O and 30.0.

(6) Frequency modulation (FM) and amplitude modulation (AM) noise. Measure-

(a) FM noise to an accuracy of ± 2 dB in the range of 40 to 80 dB below the carrier.
(b) AM noise to an accuracy of ± 3 dB in the range of 100 to 140 dB below the carrier.

c. Reference oscillator output

(1) Frequency. Measure the reference oscillator frequency to an accuracy of ± 10 Hz in the IF region.
(2) Voltage. Measure voltage output to an accuracy of ± 0.05 V, root mean square (rms) in the region of 1.0 V.

d. Built in test (BIT) functions

(1) BIT FMR Ramp. Measure ramp width to an accuracy of ± 30 microseconds(v sec.) in the region of 8 milliseconds (ms) .
(2) BIT reference. Compare TO/LO offset against BIT reference to an accuracy of ± 10 Hz in the IF region.
(3) Phase lock bandwidth. Verify bandwidth to an accuracy of ± 10 Hz over a range in the region of 4 megahertz (MHz).
(4) Verify BIT X-band switch isolation in excess of 30 dB and measure BIT target power in the region of -SO dBm, both to an accuracy of ± 2.0 dB for the following:

(a) BIT RCVR TGT
(b) BIT RADOME HORN TGT
(c) MOAT TGT

(5) Verify BIT power monitors. Measure BIT TO and continuous wave illuminator (CWI) monitor levels to an accuracy of ± 1.0V in the region of 10.5V.

e. Pulse compression. Measure incremental frequency of inverted spectrum to an accuracy of ± 10 Hz over a range in the region of 3 MHz.

f. CWI driver.

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(1) Power. Measure the CWI driver power output to an accuracy of ± 0.6 dB in the region of +17 dBm.
(2) Frequency. Measure all channel frequencies to an accuracy of - 1 kHz at X-band.
(3) Modulation. AIM-7E/7F

(a) Measure modulation frequencies of less than 1 kHz to an accuracy of ± 0.5 Hz.
(b) Measure modulation frequencies greater than 100 kHz to an accuracy of ± 25 Hz.
(c) Measure modulation indices to an accuracy of ± S percent for indices between 0.0 and 30.

(4) FM and AM noise measurement

(a) FM noise to ± 2 dB
(b) AM noise to ± 3 dB.

3.4.1.2 Synchronizer-Transmitter SN-487( )/AWG-9. The RFTS shall provide the capability for performing the following tests on the synchronizer-transmi tter.

a. BIT

(1) Measure frequency of 6 Hz Monitor signal to an accuracy of ± 0.1 Hz.
(2) Measure amplitude of 6 Hz Monitor signal to an accuracy of ± 5 percent in the region of 6 V.
(3) Measure BIT pulse repetition frequency (PRF) to an accuracy ± 1 part in 300 kHz in the region of 300 kHz.
(4) Measure the amplitude of the BIT PRF pulses to an accuracy of ± 5 percent in the region of 6 V, and the rise and fall times to an accuracy of ± 5 percent.

b. Output command signals

(1) Measure the amplitude of the output logic signals to an accuracy of ± 5 percent.
(2) Measure the rise and fall times to an accuracy of ± 5 percent.
(3) Measure the pulse repetition frequencies to an accuracy of ± 1 part in 300 kHz.
(4) Measure the pulse width of the transmitter gate signal to an accuracy of ± 100 nanoseconds (nsec) in the vicinity of 50 usec.
(5) Measure the timing of all command signals with respect to transmitter gate, or equivalent, to an accuracy of ± 100 nsec.

c. Pulse compression function tests

(1) Pulse compression gain. Measure the pulse compression gain to an accuracy of ± 1 dB.

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(2) Measure the pulse width of signals less than 3 μ sec in duration to an accuracy of ± 0.05 μ sec and of signals greater than 3.0 μ sec to an accuracy of ± 0.1 μ sec.
(3) Frequency linearity, across the compressed pulse output (side lobe level). Measure the side lobe level to an accuracy of ± 2 dB in the region of 30 dB below the compressed pulse.

3.4.1.3 Transmitter, Radar T-1224( )/AWG-9. The RFTS shall provide the capability for performing the following tests on the radar transmitter:

a. Transmitter power. Measure the transmitter average power output of the BIT coupler to an accuracy of ± 0.2 dB.
b. CWI power. Measure the CWI power output of the BIT coupler to an accuracy of ± 0.2 dB.
c. Transmitter spectrum. Verify central line power spectrum.
d. Malfunction protection. Verify operation of the following.

(1) Gridded traveling wave tube (GTWT) and CWI arc detector
(2) Low solenoid current
(3) Body current
(4) Pressure loss switch

e. Check waveguide pressurization
f. Verify operation of waveguide switches
g. Modulator assembly. Check the following:

(1) GTWT grid pulse waveform
(2) GTWT filament voltage

3.4.1.4 Power Supply PP-6770( ) /AWG-9. The RFTS shall provide the capability for performing the following tests on the collector power supply:

a. Measure output voltage under full load to an accuracy of + 1.0 percent in the region of -12.5 kiloVolts (kV).
b. Measure output ripple under full load to an accuracy of ± 10 millivolts (mV) in the region of 100 mV through the range of 400 to 4800 Hz.

3.4.1.5 Power Supply PP-6771( )/AWG-9. The RFTS shall provide the capability for performing the following tests on the beam power supply:

a. Voltage and regulation.

(1) Voltage. Measure output voltage under two load conditions to an accuracy of ± 1 percent in the region of -18 kV.
(2) Regulation. Measure the change in output voltage from no load to full load to an accuracy of ± 1.0 percent.

b. Ripple

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(1) Ripple. Measure output ripple under load to an accuracy of ± 10 mV in the region of 100 mV through the range of 400 to 4800 Hz.

3.4.1.6 Power Supply PP-6772( )/AWG-9. The RFTS shall provide the capability for performing the following tests on the solenoid power supply:

a. Current regulation. Measure current regulation to an accuracy of ± 1 percent with normal load, with shorted load, and with shorted sense in the range of 22 to 32 amperes.
b. Ripple. Measure output ripple to an accuracy of ± 3 milli-amperes (mA) through the range of 400 to 2400 Hz.

3.4.1.7 Receiver, Radar R-1785( )/AWG-9. The RFTS shall provide the capability for performing the following tests on the radar receiver:

a. Blanking isolation

(1) Verify blanking isolation in excess of 50 dB.
(2) Verify blanking logic

b. BIT Couplers. Measure coupling factor to an accuracy of ± 2 dB in the region of 40 dB.
c. Preamplifier

(1) Measure gain to an accuracy of ± 1 dB in the region of 30 dB.
(2) Measure bandwidth at the 3.0 dB points to an accuracy of ± 0.3 MHz in the region of 6.0 MHz.
(3) Measure automatic gain control (AGC) to an accuracy of ± 1.0 dB in the region of 20 dB.
(4) Measure post attenuation to an accuracy of ± 0.3 dB in the region of 12 dB.

d. Radar receiver - Paramp "ON" and "OFF"

(1) Verify bandwidth at the 1.0 dB points to an accuracy of ± 10 MHz over the receiver operating range.
(2) Measure noise figure to an accuracy oft 0.6 dB.

e. Verify control logic.

3.4.1.8 Converter, Signal Data CV-2950( )/AWG-9 and CV-3225( )/AWG-9. The RFTS shall provide the capability for performing the following tests on the doppler processor signal data converter:

a. Measure the unit gain of the clutter processor to an accuracy of ± 2.0 dB in the region of 50 dB.
b. Sub-band filter operation

(1) Measure the bandwidth to an accuracy of ± 0.5 kHz in the region of 19.2 kHz.

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(1) Measure the discriminator zero to an accuracy of ± 0.05 V in the region of zero volts.
(2) Measure the tracker test frequency to an accuracy of ± 500 Hz in the region of 7 MHz.
(3) Measure the second integrator output to an accuracy of ± 5 percent in the region of ± 10 V [track mode).

j. Tracking sensitivity (acquisition mode operation).

(1) Verify speed gate positioning by computer simulation.
(2) Measure the signal-to-noise ratio (S/N) for 50 percent fast frequency on-target (FFOT) to an accuracy of ± 3 dB.

k. Accelerating target tracking (tracking mode operation). Verify accelerating target tracking at various “G” levels to an accuracy of ± 1 “G”.

l. Polar error

(1) At the modulating frequencies, measure the polar error phase shift with continuous wave (CW) carrier to an accuracy of ± 5 percent.
(2) At the modulating frequencies, measure the polar error phase shift with noise carrier to an accuracy of ± 5 percent.
(3) Verify 4 phase potentiometer settings equivalent to phase shift to ± 1 percent.
(4) Measure polar error amplitude with 50 percent modulated carrier to an accuracy of ± 5 percent.

m. Range rate outputs

(1) Measure the target doppler RF output to an accuracy of ± 2 kHz in the region of 7.2 MHz and measure the amplitude to an accuracy of ± 6 mV rms in the region of 70 to 200 mV rms.
(2) Measure the VCO frequency output to an accuracy of ± 2 kHz in the region of 300 kHz, and measure the amplitude to an accuracy of ± 0.2 V rms in the region of O to s v rms.
(3) Verify switching logic states.

n. On-target (OT) filters

(1) Verify three OT filter states (FFOT).
(2) Verify BIT VCO slew operation.

o. BIT Log output. Verify BIT log input versus output to an accuracy of ± 5 percent.

p. Speed tracker logic tests.

(1) Measure false slam rate to an accuracy of ± 1 false alarm per second.

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(3) Verify receiver blanking operation.

d. Standard serial interface (SSI). Verify the logic level states of all SSI output signals.

3.5 Workmanship. Applicable.

3.6 Elements.

3.6.1 Console, Test Radio Frequency Transmitter OJ-247( )/AWM-23(V). The equipment of this console shall be as specified in the following.

3.6.1.1 Panel, Test, Noise Measurement ID-1860( )/AWM-23(V) (NMTP. The NMTP shall provide capability to measure frequency modulation (FM) and amplitude modulation (AM) on the X-band carrier frequencies when used in conjunction with the spectrum analyzer (3.6.2.3) .

3.6.1.2 RF power meter. An RF power meter, General Microwave 460B-4, or equivalent, shall be supplied and shall provide capability to perform the Oscillator, Radio Frequency 0-1633( )/AWG-9 and Transmitter, Radar T-1224( )/AWG-9 microwave output power measurements.

3.6.1-3 RF voltmeter. An RF voltmeter, Hewlett-Packard 3406A, or equivalent, shall be supplied and shall provide capability to perform RF voltage measurements during the Converter, Signal Data CV-2950( )/AWG-9 and CV-3225( )/AWG-9 ripple and bandwidth tests.

3.6.1.4 Panel, Coolant, Disconnect MX-9305( )/AWM-23(V). The coolant disconnect panel shall provide coolant fittings on the front panel and space behind the panel for the bay structure mounted plumbing that controls and distributes the coolant. In addition, the panel assembly shall provide diverter valve, manifold, and relief valve access.

3.6.1.5 Mounting MT-4498( )/AWM-23(V). The mounting shall provide mounting for the WRA under test and also provide the necessary electrical, air, and coolant connections to the WRA under test (see 6.9)-

3.6.1.6 Drawer, coolant drain. The coolant drain drawer shall provide the capability to drain the liquid coolant from any one of the following WRAs: Oscillator, Radio Frequency 0-1633( )/AWG-9, Transmitter, Radar T-1224( )/AWG-9, Power Supplies PP-6770( )/AWG-9 and PP-6771( )/AWG-9, and Receiver, Radar R-1785( )/AWG-9. The drawer shall be removable from the RFTS to permit disposal of the liquid coolant (see 6.9).

3.6.1.7 Panel, slave access. The slave access panel shall provide access to mounting and connections (electrical and liquid coolant) for the beam, collector, and solenoid power supplies These power supplies shall provide the high voltages and currents required for operating and testing of Transmitter, Radar T-1224( )/AWG-9.

3.6. 1.7.1 Power Supply PP-6770( )/AWG-9. A collector power supply shall be supplied and shall provide -12.5 kVdc collector power to the GTWT.

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(2) Measure FFOT to delayed velocity on target (DVOT) timing to an accuracy of ± 20.0 ms in the region of 120 ms.
(3) Measure ECLIPSE to FFOT tinting to an accuracy of ± 20.0 ms in the region of 87 ms.
(4) Measure ECLIPSE to FFOT timing to an accuracy of ± 8.0 ms in the region of 34 ms.

3.4.1.9 Converter, Signal Data, CV-2946( )/AWG-9, CV-3224( )/AWG-9 and CV-3366( )/AWG-9. The RFTS shall provide the capability for performing the following tests on the low pulse repetition frequency (PRF) signal data converter:

a Air-to-air simulation

(1) Verify air-to-air acquisition function at various ranges in both manual and automatic modes of operation.
(2) Verify air-to-air track at various ranges, range rates, and accelerations.

(a) Measure range to an accuracy of ± 10 feet plus 0.2 percent of range.
(b) Measure the amplitude of the Low PRF Video signal in the region of 5.0 V p-p to an accuracy of ± 5 percent.
(c) Measure the amplitude of the Range On Target (ROT) signal in the region of ± 4 volts to an accuracy of ± 5 percent.
(d) Measure the amplitude of the Low PRF Polar Error signal in the region of 18 V p-p to an accuracy of ± 5 percent.

b. Boresight ranging simulation

(1) Verify boresight range acquisition at various ranges and note minimum acquisition and reacquisition time.
(2) Verify boresight range tracking at various ranges, range rates, and accelerations.

(a) Measure range to an accuracy of ± 10 feet plus 0.2 percent of range.
(b) Measure the amplitude of the Low PRF Video signal in the region of 5.0 V p-p to an accuracy of ± 5 percent.
(c) Measure the amplitude of the ROT signal in the region of +4 V to an accuracy of ± 5 percent.

c. Automatic gain control (AGC)

(1) Verify the BIT AGC logic level state.
(2) Measure the amplitude of the pre-amp AGC signal in the region of 0 to -10 V to an accuracy of ± S percent.

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(3) Verify receiver blanking operation.

d. Standard serial interface (SSI). Verify the logic level states of all SSI output signals.

3.5 Workmanship. Applicable.

3.6 Elements.

3.6.1 Console, Test Radio Frequency Transmitter OJ-247( )/AWM-23(V). The equipment of this console shall be as specified in the following.

3.6.1.1 Panel, Test, Noise Measurement ID-1860( )/AWM-23(V) (NMTP). The NMTP shall provide capability to measure frequency modulation (FM) and amplitude modulation (AM) on the X-band carrier frequencies when used in conjunction with the spectrum analyzer (3.6.2.3) .

3.6.1.2 RF power meter. An RF power meter, General Microwave 460B-4, or equivalent, shall be supplied and shall provide capability to perform the Oscillator, Radio Frequency 0-1633( )/AWG-9 and Transmitter, Radar T-1224( )/AWG-9 microwave output power measurements.

3.6.1.3 RF voltmeter. An RF voltmeter, Hewlett-Packard 3406A, or equivalent, shall be supplied and shall provide capability to perform RF voltage measurements during the Converter, Signal Data CV-2950( )/AWG-9 and CV-3225( )/AWG-9 ripple and bandwidth tests.

3.6.1.4 Panel, Coolant, Disconnect MX-9305( )/AWM-23(V). The coolant disconnect panel shall provide coolant fittings on the front panel and space behind the panel for the bay structure mounted plumbing that controls and distributes the coolant. In addition, the panel assembly shall provide diverter valve, manifold, and relief valve access.

3-6.1.5 Mounting KT-4498( )/AWM-23(V). The mounting shall provide mounting for the WRA under test and also provide the necessary electrical, air, and coolant connections to the WRA under test (see 6.9).

3.6.1.6 Drawer, coolant drain. The coolant drain drawer shall provide the capability to drain the liquid coolant from any one of the following WRAs. Oscillator, Radio Frequency 0-1633( )/AWG-9, Transmitter, Radar T-1224( )/AWG-9, Power Supplies PP-6770( )/AWG-9 and PP-6771( )/AWG-9, and Receiver, Radar R-1785( )/AWG-9. The drawer shall be removable from the RFTS to permit disposal of the liquid coolant (see 6.9).

3.6.1.7 Panel, slave access. The slave access panel shall provide access to mounting and connections (electrical and liquid coolant) for the beam, collector, and solenoid power supplies. These power supplies shall provide the high voltages and currents required for operating and testing of Transmitter, Radar T-1224( )/AWG-9.

3.6.1.7.1 Power Supply PP-6770( )/AWG-9. A collector power supply shall be supplied and shall provide -12.5 kVdc collector power to the GTWT.

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3.6.1.7.2 Power Supply PP-6771( )/AWG-9. A beam power supply shall be supplied and shall provide regulated -18 kVdc beam voltage to Transmitter, Radar T-1224( )/AWG-9.

3.6.1.7.3 Power Supply PP-6772( )/AWG-9. A solenoid current power supply shall be supplied and shall provide regulated solenoid current to Transmitter, Radar T-1224( )/AWG-9.

3.6.2 Console, Test Radio Frequency, Receiver OJ-248( )/AWM-23(V). The equipment of this console shall be as specified below.

3.5.2.1 Microwave swept oscillator. A microwave swept oscillator, E-H Research Labs, Inc., Model 574-56, or equivalent, shall be supplied and shall provide capability to perform the Receiver, Radar R-1785( )/AWG-9 paramp test and alignment. The swept oscillator shall also provide inputs to Receiver, Radar R-1785( )/AWG-9 to perform the noise figure measurements.

3.6.2.2 Noise figure meter. A noise figure meter, General Microwave 551A-27, or equivalent, shall be supplied and shall provide capability to perform Receiver, Radar R-1785( )/AWG-9 noise figure measurements.

3.6.2.3 Spectrum analyzer. A spectrum analyzer, Hewlett-Packard 141T/ 8552B/8553B with plug-in module 8555A and 8556A, or equivalent, shall be supplied. The spectrum analyzer shall provide the capability to perform Power Supplies PP-6770( )/AWG-9, PP-6771(.)/AWG-9, and PP-6772( )/AWG-9 low frequency noise measurement; video modulation analysis in conjunction with the noise measurement test panel (3.6.1.1); Converter, Signal Data CV-2950( )/ AWG-9 and CV-3225( )/AWG-9 crystal filter tests; analyze the Synchronizer-Transmitter SN-487( )/AWG-9 spectrum; and perform X-band measurements.

3.6.2.4 Switching Unit, Fault Isolation SA-1893( )/AWM-23(V). The fault. isolation switching unit shall provide capability to perform the fault isolation signal switching during fault isolation tests of the WRAs and during test station fault isolation.

3.6.2.5 Test Set, Radio Frequency Power TS-3276( )/AWM-23(V). The radio frequency power test set shall provide microwave switching, attenuation, termination and interconnection capabilities to connect the Oscillator, Radio Frequency 0-1633( )/AWG-9, Transmitter, Radar T-1224( )/AWG-9, and Receiver, Radar R-1785( )/AWG-9 WRAs to the RFTS for test purposes. This drawer shall contain a crystal mixer for use in FMR linearity measurements and another crystal mixer for modulation analysis.

3.6.2.5.1 Oscillator, Radar Frequency 0-1633( )/AWG-9. A radar frequency oscillator shall be supplied and shall provide a precision RF source for the RFTS and WRAs under test.

3.6.3 Console, Test, Radio Frequency? Receiver-Transnutter OJ-249( )/ AWM-23(V). The equipment of this console shall be as specified in the following.

3.6.3.1 Oscilloscope. An oscilloscope, Hewlett-Packard 181 AR with plug-in amplifier 1801A-003 and plug-in time base and delay generator 1821A, or equivalent, shall be supplied and shall provide display of waveforms.

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3.6.3.2 Panel, Test, Electrical C-9177( )/AWM-23(V). The electrical test panel shall provide the controls, displays and switching capability to conduct functional and fault isolation tests on the WRAs assigned to the RFTS. Controls on the front panel shall be provided to command WRA Power On or Power Off, command emergency off for station power, select and enable functional or fault isolation tests, select and apply input signals to WRAs, select measurement instruments during test, command SSI Word Generator to provide desired word, control setting of attenuators, and control level of simulated NRA inputs. Displays on the front panel shall be provided to display test number, test mode and status, and SSI BIT fault.

3.6.3.3 Panel, access. The access panel shall provide cover for bay structure mounted electrical wiring.

3.6.3.4 Comparator, Signal CM-434( )/AW-23(V). The signal comparator shall provide:

a. The capability to determine modulation frequencies and modulation indices associated with the AIM-7E and AIM-7F Missiles.
b. Timing and gating functions necessary for operation and test of the receiver-transmitter group WRAs.
c. The capability to measure the slope and linearity of the FMR ramp modulated X-band output of the Oscillator, Radio Frequency 0-1633( )/AWG-9 under test.

3.6.3.5 Dummy Load, Electrical DA-630( )/AWM-23(V), The dummy load shall provide the capability to test the high power outputs of the transmitter power supplies by means of operating into a liquid cooled load assembly. Resistive load elements and associated high voltage switching relays shall be enclosed within the liquid filled assembly.

3.6.3.6 Transducer, TR-283( )/AWM-23(V). The transducer shall provide the capability to provide fault and monitoring signals for detection of inadequate cooling by means of coolant flow transducers.

3.6.4 Console, Test, Radio Frequency OJ-250( )/AWM-23(V). The equipment of this console shall be as specified below.

3.6.4.1 Panel, Power Distribution SB-3694( )/AWM-23(V). The power distribution panel shall provide the capability to:

a. Control RFTS power function.
b. Indicate that input power forms are applied to the test station.
c. Provide overload protection for the RFTS.
d. Display RFTS malfunction indication.

3.6.4.2 Signal generator. The signal generator, RF Communications Incorporated RF-808-863-008, or equivalent, shall provide IF signals.

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3.6.4.3 Counter/timer. A counter/timer, Hewlett-Packard 5326B/001/011, or equivalent, shall be supplied and shall provide capability to perform the Synchronizer-Transmitter SN-487( )/AWG-9 and Converter, Signal Data CV-2946( )/AWG-9, CV-3224( )/AWG-9 and CV-3366( )/AWG-9 time interval measurements.

3.6.4.4 Frequency counter. A frequency counter, Systron-Dormer 6316A-29, or equivalent, shall be supplied and shall provide capability to perform frequency measurements (primarily X-band frequencies) .

3.6.4.5 Digital multimeter. A digital multimeter, Systron-Dormer 9340-5B, or equivalent, shall be supplied and shall provide capability to perform ac and dc voltage measurements, and resistance measurements.

3.6.4.6 Frequency response analyzer. A frequency response analyzer, EMR Hatboro 1410 AR-1, or equivalent, shall be supplied and shall provide capability to perform the Converter, Signal Data CV-2946( )/AWG-9, CV-3224( )/AWG-9, CV-2950( )/AWG-9, CV-3225( )/AWG-9 and CV-3366( )/AWG-9 polar error phase shift measurements.

3.6.4.7 Range simulator. The range simulator, Instrumentation Technology Corporation 9007, or equivalent, shall be supplied and shall provide capability to perform the Converter, Signal Data CV-2946( )/AWG-9, CV-3224( )/AWG-9 and CV-3366( )/AWG-9 dynamic tracking tests.

3.6.4.8 Generator, Reference Signal SG-1014( )/AWM-23(V). The reference signal generator shall provide the following capabilities:

a. Comparison of signals from the speed tracker portion of the Converter, Signal Data CV-2950( )/AWG-9 and CV-3225( )/AWG-9 to provide a GO/NO GO output.
b. Processing pulse and frequency modulated signals at the radar IF center frequency and inversion of the frequency domain spectrum along the frequency axis. The IF spectrum inverter shall be used to test Synchronizer-Transmitter SN-487( )/AWG-9. In addition, a pulse delay capability shall be included.

3,6.4.9 Power Supply PP-6848( )/AWM-23(V). The power supply shall provide the capability to supply dc voltages to WRAs and RFTS circuits, using a single dc power supply as a source (3.6.4.10]. The power supply shall also provide the capability to sense the output power forms for overvoltage, undervoltage, and overcurrent for RFTS and WRA protection as well as for continuous monitor capability.

3.6.4.10 power Supply, dc. The dc power supply, Lambda LB706-FM-3632, or equivalent, shall provide the capability to supply 200 Vdc at 10 amperes to Power Supply PP-6848( )/AWM-23(V).

3.6.5 Accessories - RFTS. The RFTS accessories shall be composed of items not permanently attached to the RFTS.

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3.6.5.1 Mounting adapters. The mounting adapters shall provide WRA mounting, liquid coolant connections, air cooling connections, and connection to the RFTS interface at the mounting surface.

3.6.5.1.1 Mounting adapter - modulator. The modulator mounting adapter shall serve as a holding fixture which mates with the mounting (3.6.1.5) and shall provide air cooling for the modulator assembly of Transmitter, Radar T-1224( )/AWG-9.

3.6.5.1.2 Mounting adapter- RMO and receiver. The RMO and receiver mounting adapter shall serve as a holding fixture which mates with the mounting (3.6.1.5) and shall provide liquid coolant connections for Oscillator, Radio Frequency 0-1633( )/AWG-9 and Receiver, Radar R-1785( )/AWG-9.

3.6.5.1.3 Mounting adapter - power supply. The power supply mounting adapter shall serve as a holding fixture which mates with the mounting (3.6.1.5) and shall provide liquid coolant connections for Power Supplies PP-6770( )/AWG-9 and PP-6771( )/AWG-9.

3.6.5.2 Interface waveguides. The interface waveguide/coaxial cable shall be used to provide connections between the RFTS and the following WRAs during tests.

a. Oscillator, Radio Frequency 0-1633( )/AWG-9.
b. Transmitter, Radar T-1224( )/AWG-9.
c. Receiver, Radar R-1785( )/AWG-9.

3.6.5.3 Extenders - RFTS. The RFTS shall include extender cards and cables to extend modules from their respective WRAs or GSE assemblies.

3.6.5.4 Commercial items, auxiliary. Auxiliary commercial items shall provide supplementary test capability to the commercial test equipment of the RFTS.

3.6.6 Elapsed time meter. An elapsed time meter, conforming to MIL-M-7793, shall be provided. The meter shall be readily visible during operation of the RFTS and shall be capable of registering at least 9,999 hours.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Applicable.

4.2 General. Applicable.

4.2.1 Classification. Applicable.

4.2.2 Individual items. Applicable.

4.2.3 In-process inspection. Applicable.

4.2.4 In-process data. Applicable.

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4.2.5 Acceptance test plan. Applicable. Table V shall apply.

4.2.6 Test results. Applicable.

4.2.7 Acceptance. Applicable.

4.2.8 Utilities. Applicable.

4.2.9 Configuration. Applicable.

4.3 Tests.

TABLE V

ACCEPTANCE TEST PLAN FOR RADIO FREQUENCY TEST STATION
OJ-213( )/AWM-23(V)

1.0 TEST OBJECTIVE. To demonstrate self-test and operational run-in tests of the Radio Frequency Test Station.

2.0 TEST CONDITIONS

a. AWG-9 Slave Units

(1) 481001-1XX
(2) 481013-1XX
(3) 481014-1XX
(4) 481015-1XX

b. Utilities Required (Information only, ref. 4.2.8)

(1) Power as specified by 3.2.2.1
(2) Liquid cooling as specified by 3.2.2.4
(3) Clean air cooling as specified by 3.2.2.2
(4) Nitrogen as specified by 3.2.2.6

3.0 TESTS. In accordance with Acceptance Test Procedure 3282204.

a. Perform self-test.
b. Perform operational run-in-tests.

4.3.1 Electrical and electronic items - operational run-in test. Applicable.

4.3.2 Mechanical items. Applicable.

4.4 Environmental test conditions. Applicable.

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4.5 Test location. Applicable.

4.6 Acceptance schedule. Applicable.

4.7 Test procedures. Applicable.

4.8 Rejection and retest. Applicable.

5. PREPARATION FOR DELIVERY

5.1 General. Applicable.

6. NOTES

6.1 Intended use. The RFTS is intended for intermediate level use to functionally test, fault isolate, and verify the serviceability of certain weapon replaceable assemblies (WRA) of the radar subsystem of the Weapon Control System (WCS) AN/AWG-9.

6.2 Ordering data. Applicable.

6.3 Definitions. Applicable.

6.4 Nomenclature cross reference. The following is a cross reference between the official Navy nomenclature and the major contractor part numbers. The contractor part number suffix is controlled by the configuration list in the contract.

	Official Nomenclature	Contractor Part No.
a.	Oscillator, Radio Frequency 0-1633( )/AWG-9	481001-1XX
b.	Synchronizer-Transmitter SN-487( )/AWG-9	481010-1XX
c.	Transmitter, Radar T-1224( )/AWG-9	481011-1XX
d.	Power Supply PP-6770( )/AWG-9	481013-1XX
e.	Power Supply PP-6771( )/AWG-9	481014-1XX
f.	Power Supply PP-6772( )/AWG-9	481015-1XX
g.	Receiver, Radar R-1785( )/AWG-9	481022-1XX
h.	Converter, Signal Data CV-2950( )/AWG-9 and CV-3255( )/AWG-9	481039-1XX
i.	Converter, Signal Data CV-2946( )/AWG-9, CV-3224( )/AWG-9 and CV-3366( )/AWG-9	481083-1XX
j.	Test Station, Radio Frequency OJ-213( )/AWM-23(V)	493150-1XX
h.	Console, Test, Radio Frequency Transmitter OJ-247( )/AWM-23(V)	493151-1XX

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l.	Console, Test, Radio Frequency Receiver OJ-248( )/AWM-23(V)	493152-1XX
m.	Console, Test, Radio Frequency Receiver-Transmitter OJ-249( )/AWM-23(V)	493153-1XX
n.	Console, Test, Radio Frequency OJ-250( )/AWM-23(V)	493154-1XX
o.	Kit, Tool, Special TK-222( )/AWM-23(V)	493601-1XX
p.	Cooler, Liquid, Electronic Equipment HD-957( )/AWM-23(V)	493711-1XX

6.5 Liquid cooler. Applicable.

6.6 Use of this specification. This specification is a detail specification which uses MIL-T-81734(AS) as a general specification. Where the corresponding paragraphs of MIL-T-81785 are the same as MIL-T-81734(AS), the word “applicable” is used in this specification. When the term “not applicable” is used in this specification, it means that the corresponding paragraph requirements of MIL-T-81734(AS) are not required to be met (see 6.6 of MIL-T-81734(AS)).

6.7 Abbreviations and acronyms. Applicable. In addition the following shall apply:

AM	Amplitude modulation
CTE	Comercial test equipment
CWI	Continuous wave illuminator
dBm	Decibel referred to 1 milliwatt level
DVOT	Delayed velocity on target
FFOT	Fast frequency on target
FM	Frequency modulation
GTWT	Gridded traveling wave tube
IF	Intermediate frequency
kHz	Kilohertz
kV	Kilovolts
Lo	Local oscillator
mHz	Megahertz
ms	Milliseconds
µsec	Microseconds
mV	Millivolts
ns	Nanoseconds
OT	On target
PRF	Pulse repetition frequency
RF	Radio frequency
RFTS	Radio frequency test station
ROT	Range on target
S/N	Signal-to-noise
SSI	Standard serial interface
TO	Transmitter oscillator
VCO	Voltage controlled oscillator

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6.8 Convenience outlet power. Applicable.

6.9 Bay No. 1 mounting and drawer. The mounting and drawer are not included in the assembly drawing for Bay No. 1 since the mounting extends across both Bays 1 and 2 and the drawer is supported by the mounting. They are included here for convenience-

6.10 Changes from previous issue are not marked.

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DOCUMENT IDENTIFIER AND TITLE

MIL-T-81785C(AS) Test Station, Radio Frequency OJ-213( )/AWM-23(V)

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO
	DATE

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